



Security and Governance

For data and AI on Databricks



Welcome



Rosalie Noël

Solutions Architect, Sydney

Your host today

- ~3 years with Databricks
- 6 years in the Data and AI Industry, after a previous career in Financial Risk Management
- Passionate about Data and Governance

Housekeeping

- This presentation is being recorded, we will share the recording and other materials after the session, within 48 hours
- There is no hands-on component but you will see a demo of Unity Catalog
- Use the Q&A function to ask questions. If we do not answer your question during the event, we will follow-up with you afterwards to get you the information you need
- Please fill out the survey at the end of the session so that we can improve our future sessions



Agenda

Data and AI security overview

Introduction to Unity Catalog

Key concepts and capabilities

- Centralised metadata and access controls
- Query federation & Volumes
- Sharing and Collaboration
- Discover your data with search and lineage
- Audit and monitor your data

Upgrading to Unity Catalog

Demo!

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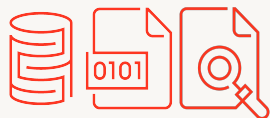
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Data and AI security

Why is it important?



Data gives a competitive advantage

Data is one of your enterprise's most critical asset. When it is protected, governed and trustworthy, it drives value creation.



Preventing unwanted risks

Data loss or misuse can have severe consequences, such as regulatory breach, reputational damage and financial losses.

Defining Data and AI Security

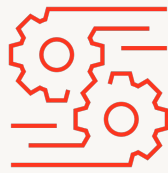
Key Principles



Confidentiality

Sensitive information is accessible only to those with appropriate permissions

- Data classification
- RBAC/ ABAC
- Data Masking



Integrity

Data remains accurate and trustworthy

- Audit and Logging
- Data Monitoring
- Single Source of Truth

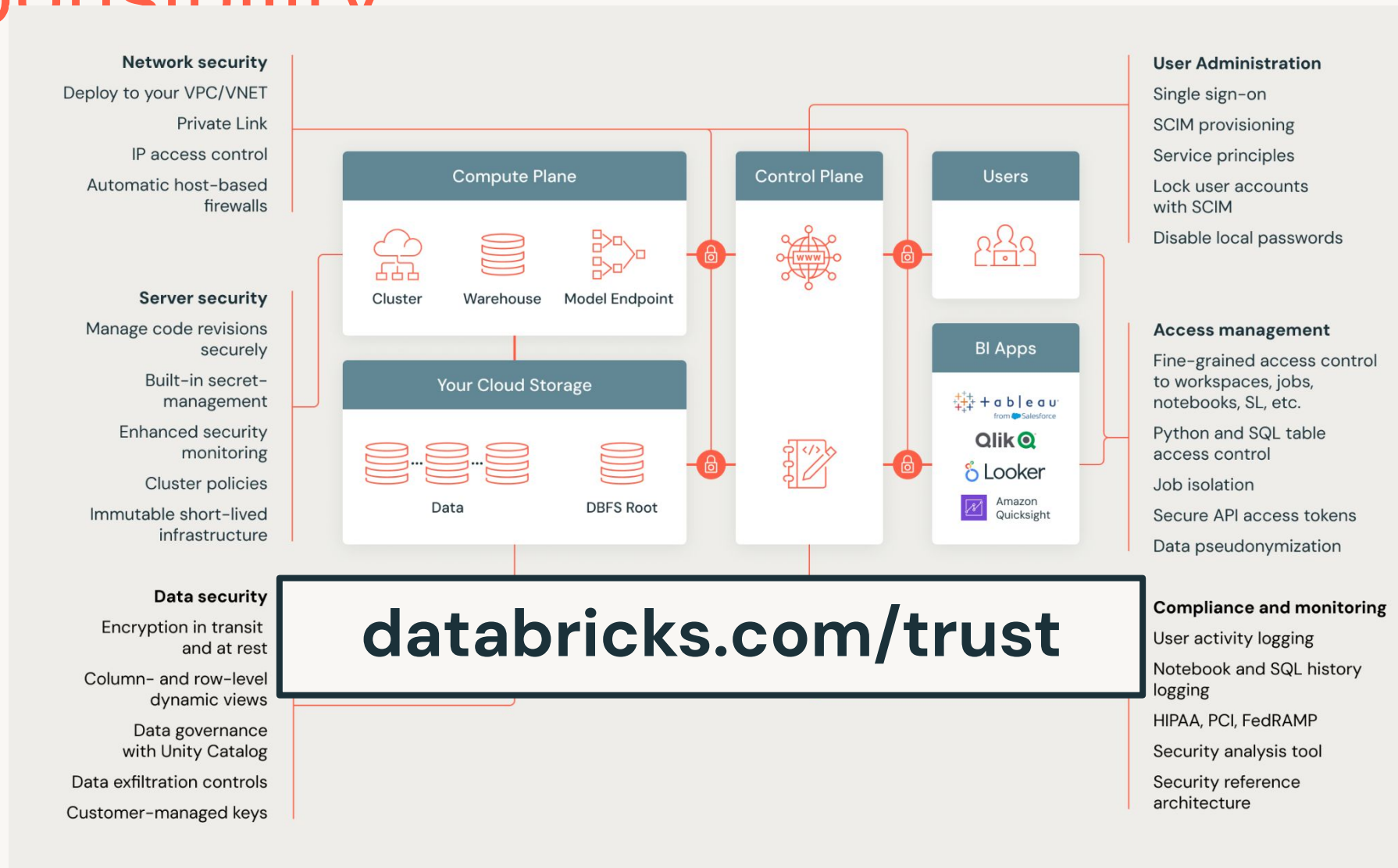


Availability

Authorised users can access data and resources without disruption

- Users Authentication
- Access Controls
- HA/DR

Databricks security is a **Shared Responsibility**



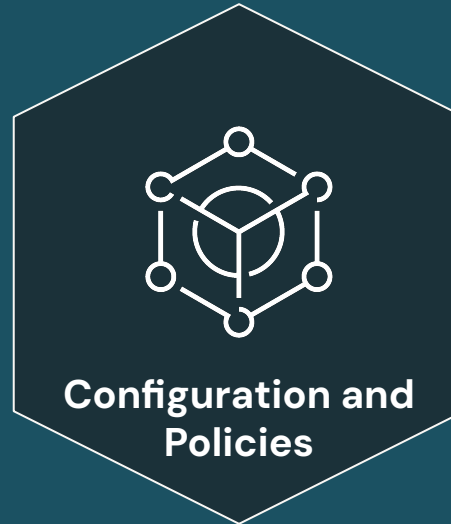
Steps to Secure Databricks Deployments

Define Security Controls



**White Papers
Best practices**

Deploy Security Controls



**Terraform
Security Reference
Architecture**

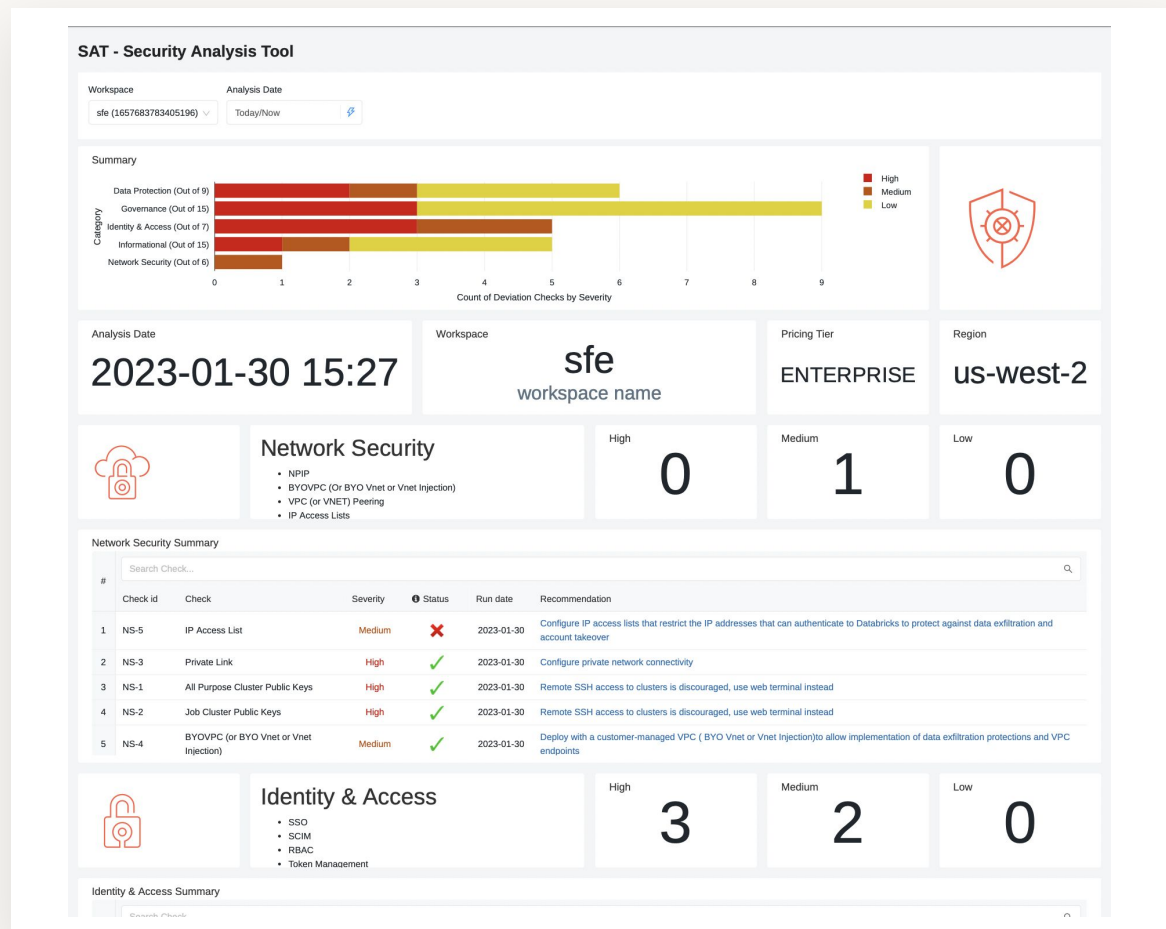
Monitor Security Controls



**Security
Assessment Tool**

Monitor: Security Analysis Tool

Monitor the security health of your account workspaces over time



- Compare workspace configurations against specific best practices
- Flag deviations for your workspaces over a period of time
- Easily identify mitigation references
- Available for AWS, Azure and GCP (including Terraform deployments)

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Two seemingly **opposite** business imperatives

Data Security

VS.

**Data
Democratisation**

**Historically, two
conflicting objectives
competing with each
other**



Cloud security



Access controls



Compliance

Open collaboration



Data domains



Immediate data access



Data Governance empowers the business

Applying the “Goldilocks Principle”

Data Security



Cloud security



Access controls



Compliance



“Just Right”

**Unity Catalog lets you have the
best of both worlds!**

**Data
Democratisation**

Open collaboration



Data domains



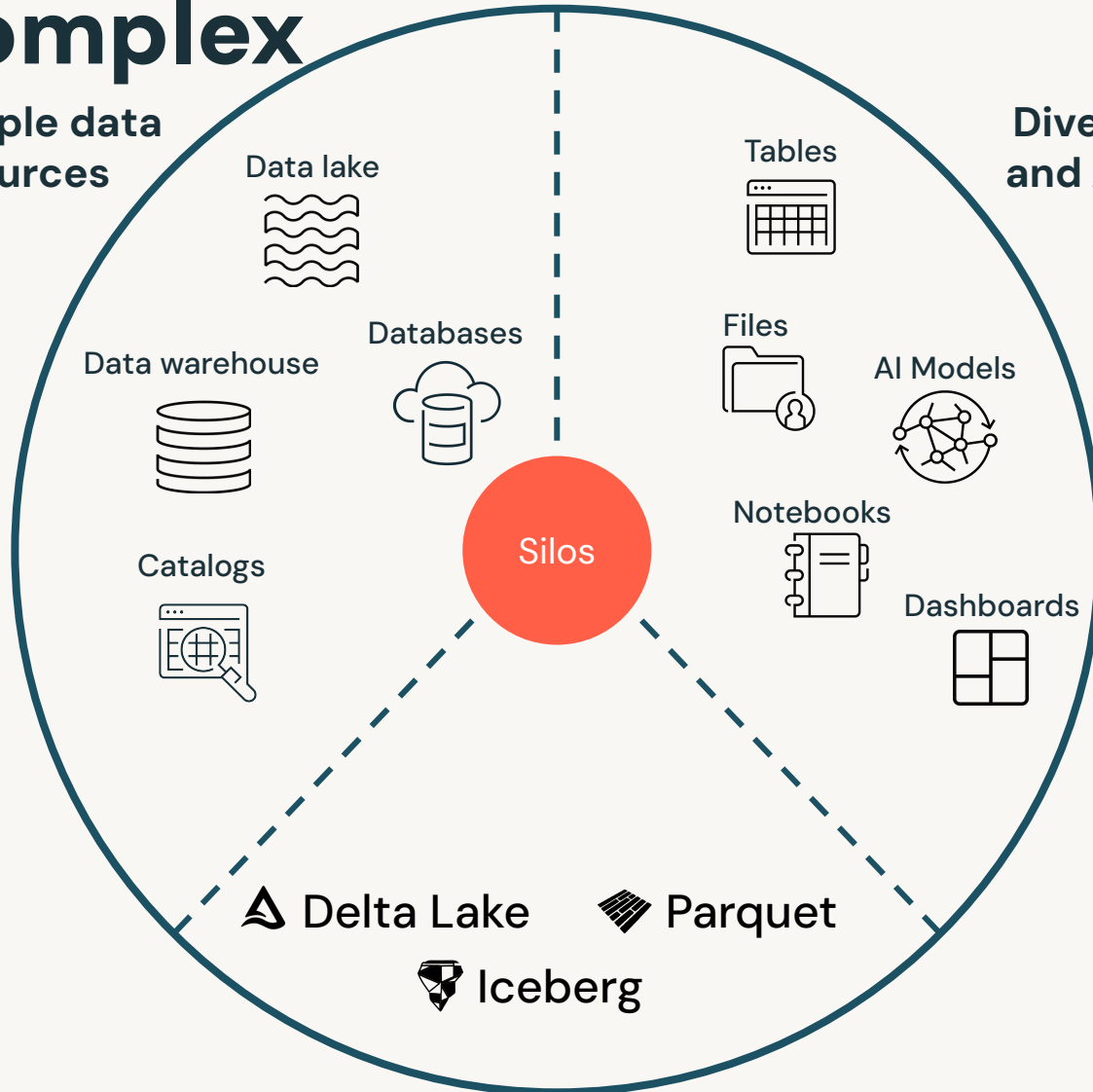
Immediate data access



Governance of **the entire data estate** is complex

Multiple data
Sources

Diverse data
and AI assets



Heterogeneous data formats



Fragmented governance for data and AI
Across access management, auditing,
monitoring, observability, and lineage

Lack of open connectivity
For access and cross-platform collaboration

Lack of built-in intelligence
For data discovery, understanding,
and gaining domain insights

These challenges **impact business value**

Fragmented governance



Compliance, security and quality risks
Operational inefficiencies

Lack of open connectivity



Vendor lock-in
Higher operational cost

Lack of built-in intelligence



Slow speed of innovation
Missed competitive edge



An **ideal governance** solution needs:

Unified Governance

All data and AI assets
and data formats

Open Connectivity

Any data source, compute
engine, and client

Built-in Intelligence

Powering domain knowledge
and platform optimizations



Unified and open governance for all data + AI



Traditional catalogs

Security

Access Control

Auditing

Collaboration

Discovery

Secure Data Sharing

Quality

Lineage

Quality Monitoring

Management

Cost Controls

Business Semantics

Tables



AI Models



Files



Notebooks



Dashboards



Delta Lake



Parquet



Iceberg

Connect to any external data source



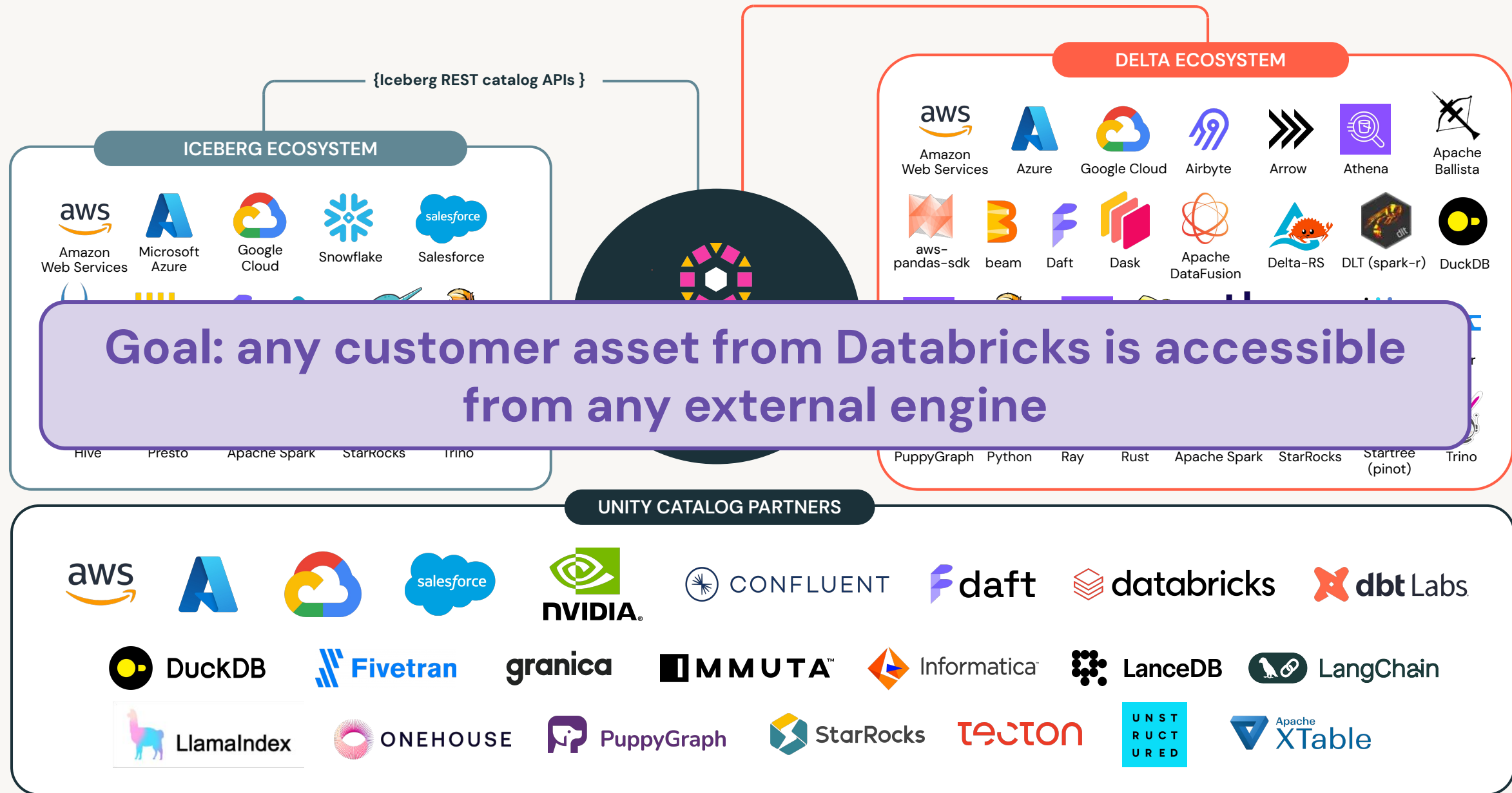
Open access and collaboration with any tool, engine or platform

Open access

- Access and query **external databases**, data warehouses, and open table format catalogs **without data replication**, reducing costs
- Avoid vendor lock-in with **accessibility from any tool**, engine or platform for broad interoperability
- **Future proof** your governance with an **open source foundation**



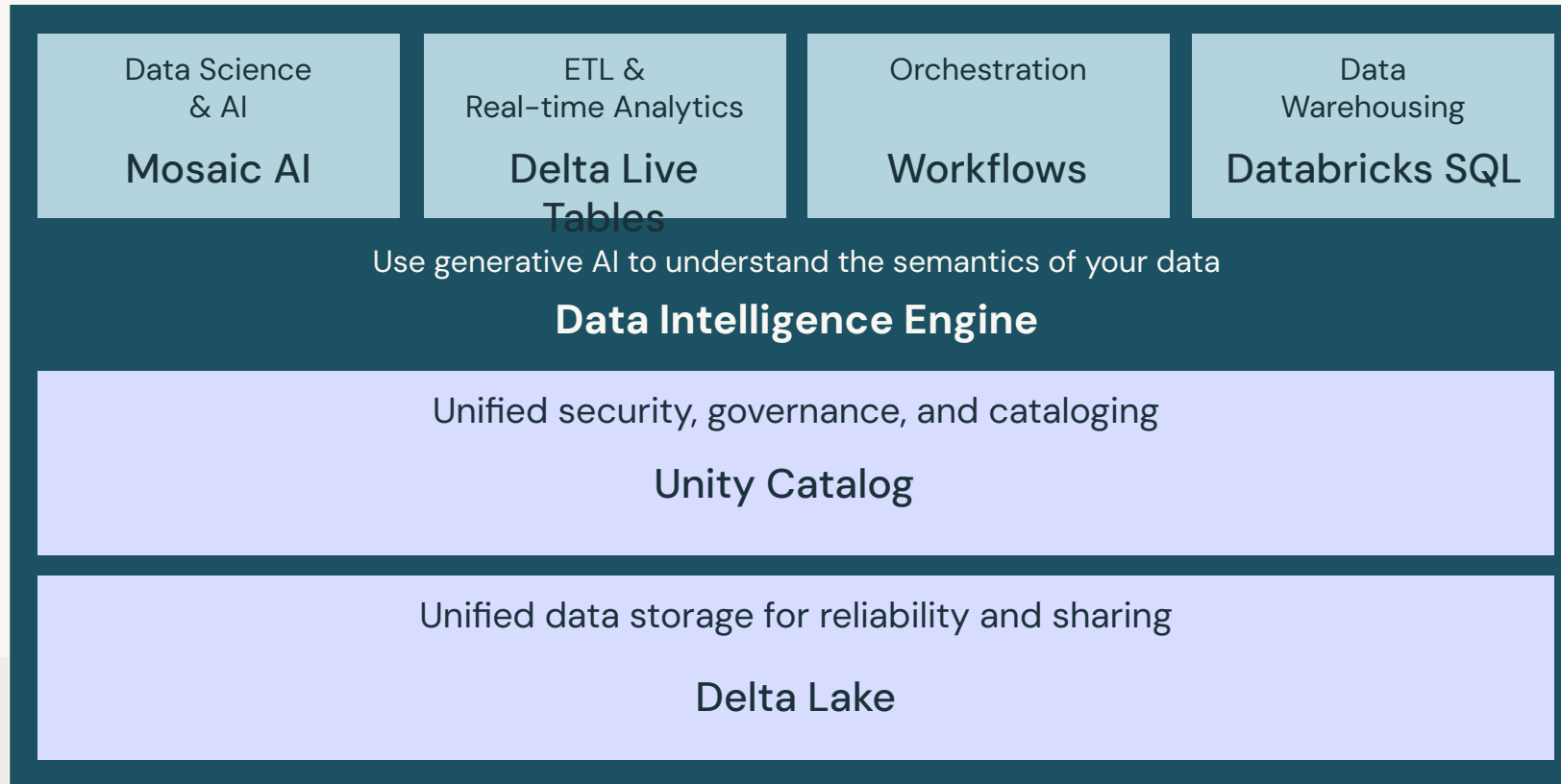
UC OSS interoperability



However, it is not just a
catalog ... in Databricks



Unity Catalog Powers Databricks Data Intelligence Platform



Open Data Lake

All Raw Data
(Logs, Texts, Audio, Video, Images)

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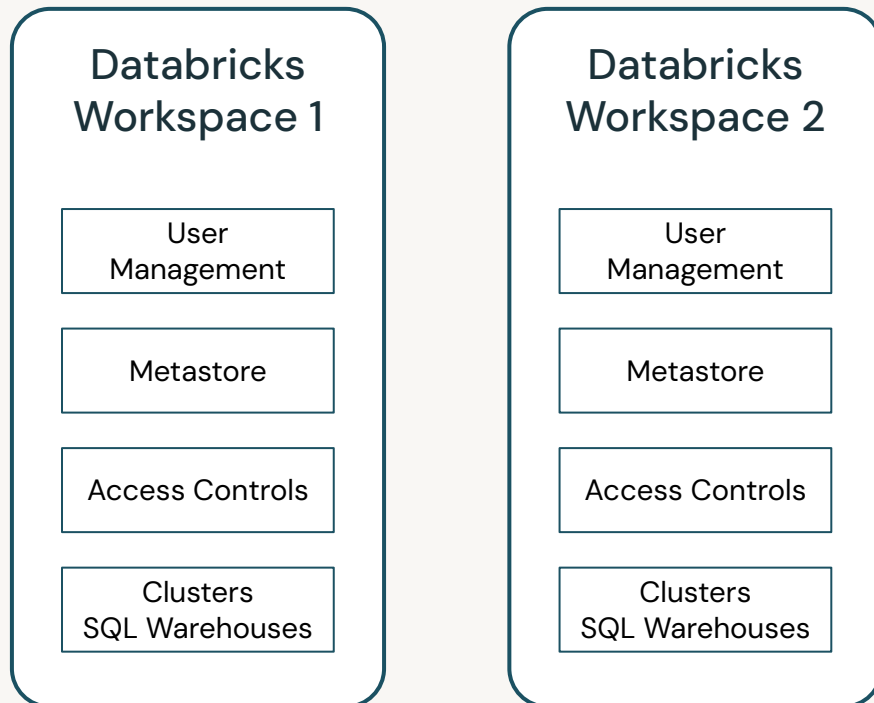
Upgrading to Unity Catalog

Demo!

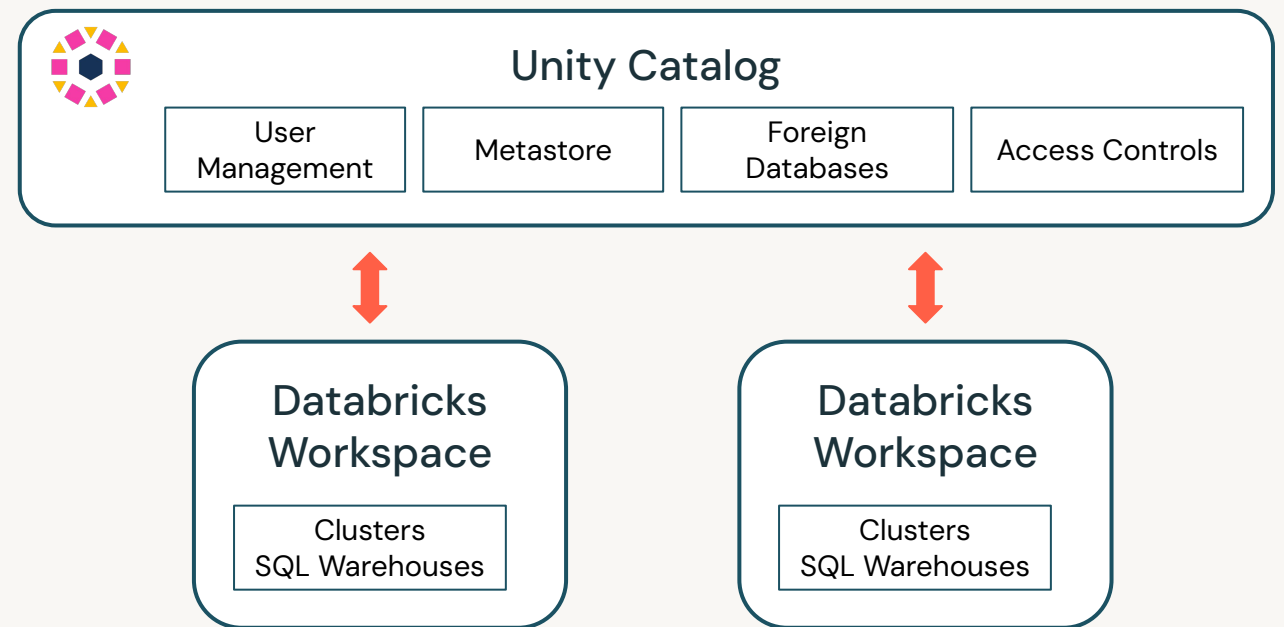
From legacy Hive Metastore to Unity Catalog

One metadata layer across file and database sources **superpowers** governance

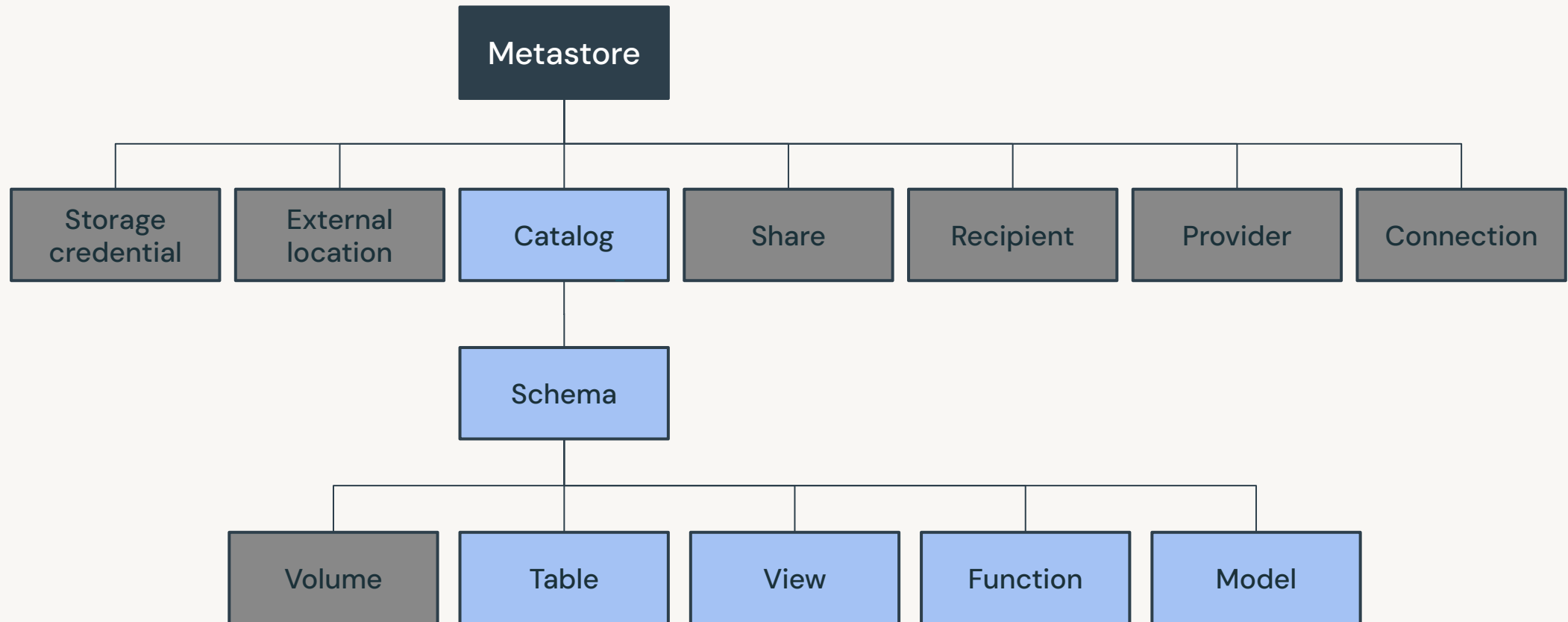
Without Unity Catalog



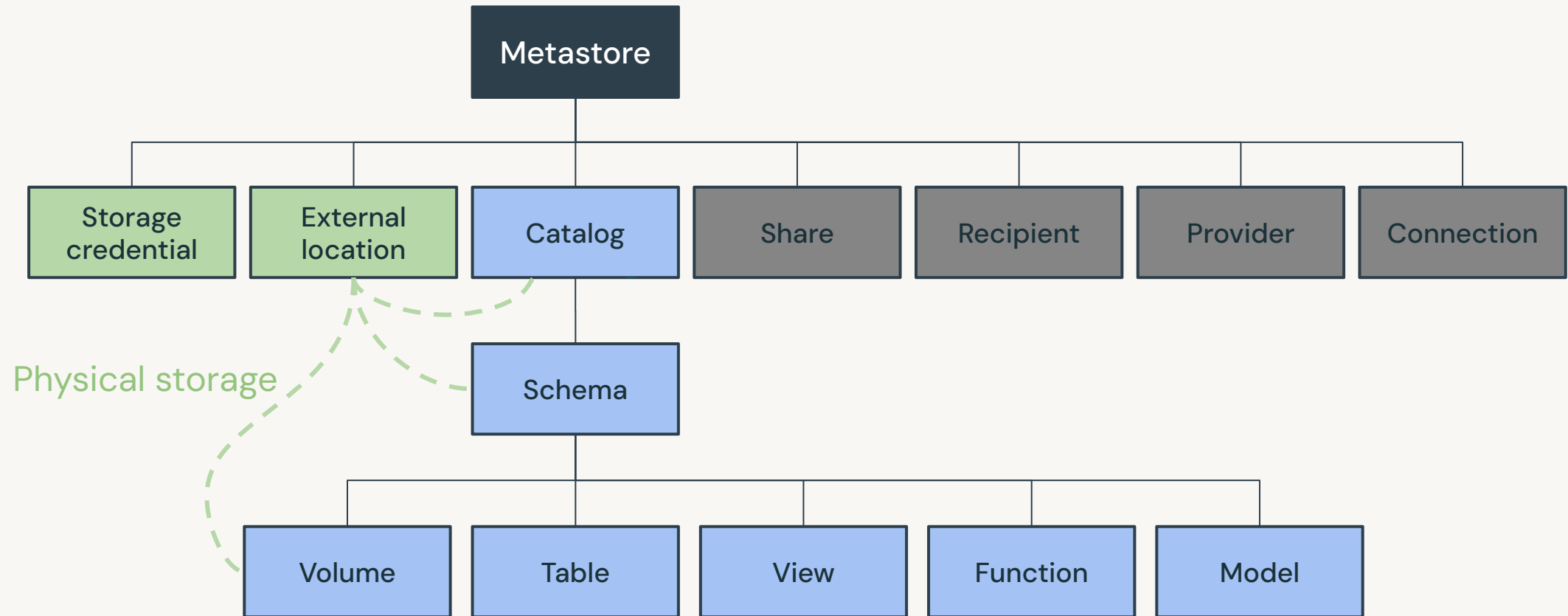
With Unity Catalog



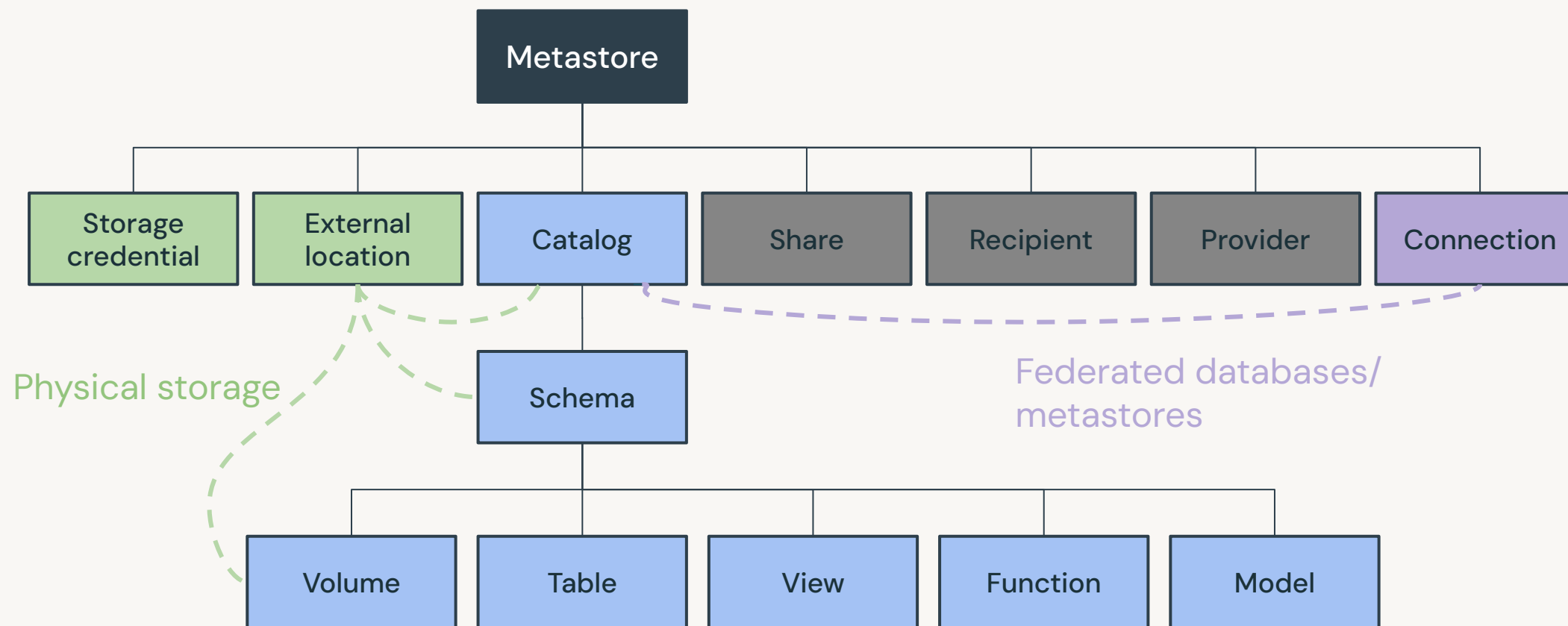
Unity Catalog Securable Objects



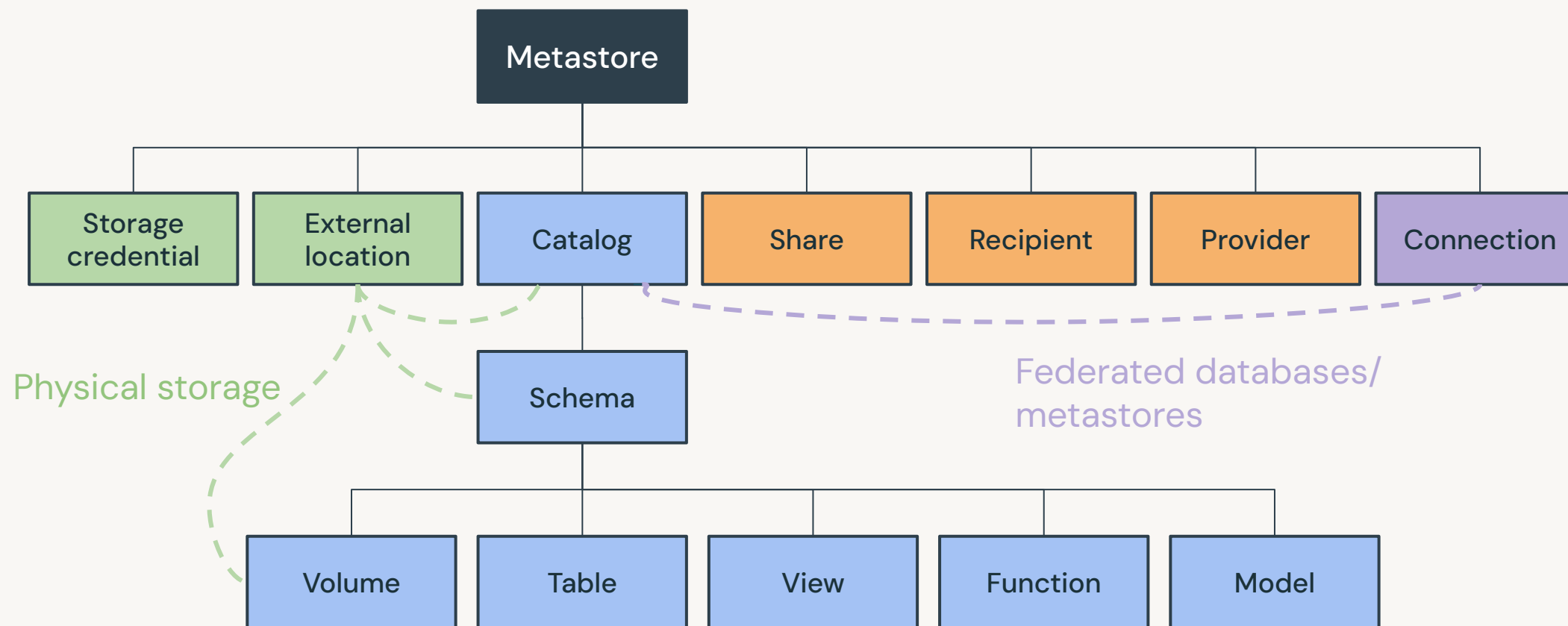
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Unity Catalog Securable Objects

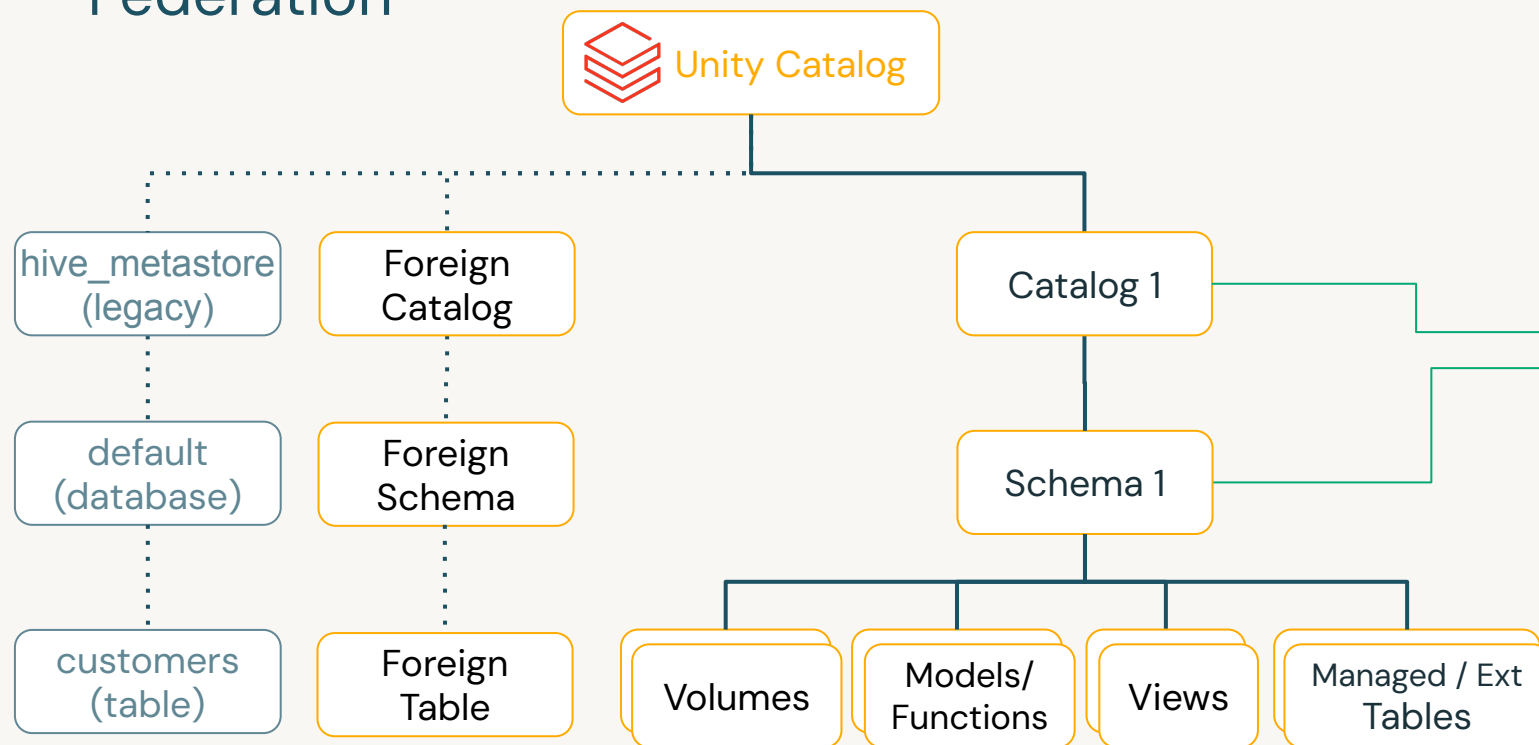


Unity Catalog Securable Objects



Governed namespace across file and database sources

Access legacy metastore and foreign databases powered by Lakehouse Federation



Data Explorer dais-gov

Data

Type to filter

- retail_dev
 - retail_prod
 - churn
 - Tables
 - churn_feature_store
 - churn_feature_store_ty
 - churn_joined
 - churn_joined_v
 - feature_table
 - training_dataset
 - user_churn_analysis
 - user_features
 - Volumes
 - Functions
 - Models
 - churn_bronze

Catalogs > retail_prod > churn > retail_prod.churn.user_features

Tags: Add

Owner: kasey.uhlenhuth@databricks.com

Columns Sample Data Details Permissions

Filter columns...

Column	Type
user_id	string
email	string
creation_date	timestamp
last_activity_date	timestamp
firstname	string
lastname	string

```
SELECT * FROM main.paul.red_wine; -- <catalog>.<database>.<table>
```

```
SELECT * FROM hive_metastore.default.customers;
```

```
SELECT * FROM snowflake_warehouse.some_schema.some_table;
```

Centralized Access Controls

Centrally grant and manage access permissions across workloads and foreign databases

Using ANSI SQL DCL

```
GRANT <privilege> ON <securable_type>  
<securable_name> TO <principal>
```

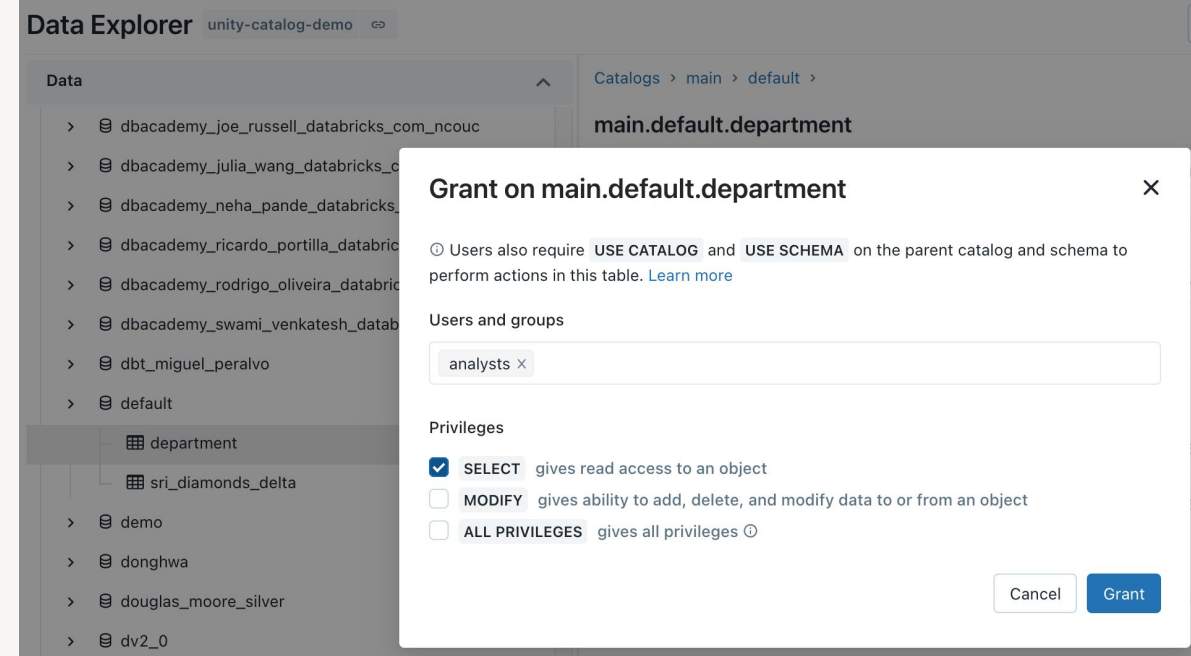
```
GRANT SELECT ON iot.events TO engineers
```

Choose
permission level

'Table' = collection of
files in S3/ADLS

Sync groups from
your identity
provider

Using UI



Row Level Security and Column Level Masking

Provide differential fine grained access to file based datasets and foreign tables

Only show specific rows

```
CREATE FUNCTION <name> ( <parameter_name >  
<parameter_type> .. )  
RETURN {filter clause whose output must be a boolean}
```

```
CREATE FUNCTION us_filter(region STRING)  
RETURN IF(IS_ACCOUNT_GROUP_MEMBER('admin'), true,  
region="US");  
  
ALTER TABLE sales SET ROW FILTER us_filter ON region;
```

Test for group
membership

Assign reusable
filter to table

Specify filter
predicates

Mask or redact sensitive columns

```
CREATE FUNCTION <name> (<parameter_name>,  
<parameter_type>, [, <column>...])  
RETURN {expression with the same type as the first  
parameter}
```

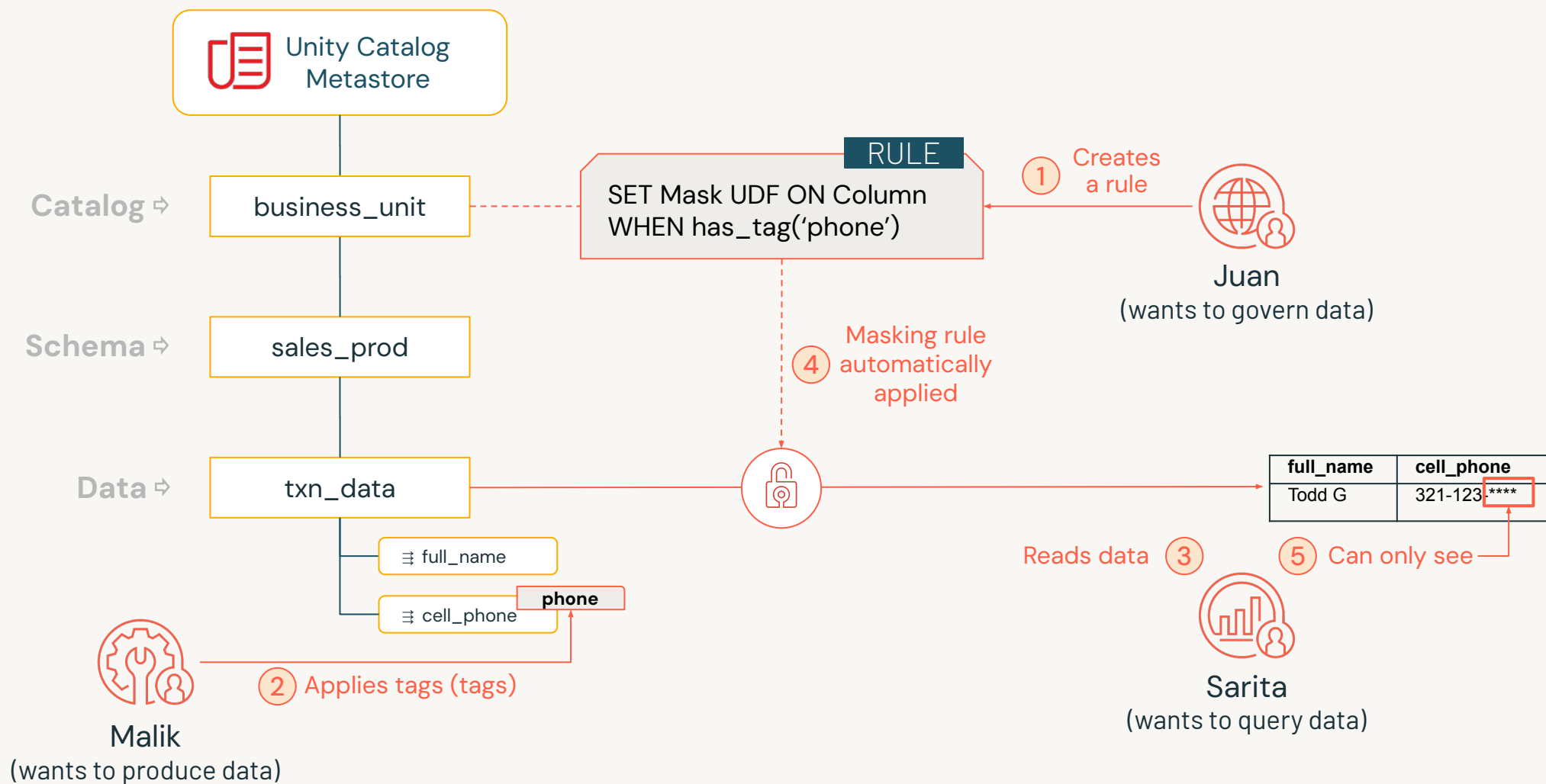
```
CREATE FUNCTION ssn_mask(ssn STRING)  
RETURN IF(IS_ACCOUNT_GROUP_MEMBER('admin'), ssn, "*****");  
  
ALTER TABLE users ALTER COLUMN table_ssn SET MASK  
ssn_mask;
```

Test for group
membership

Assign reusable
mask to column

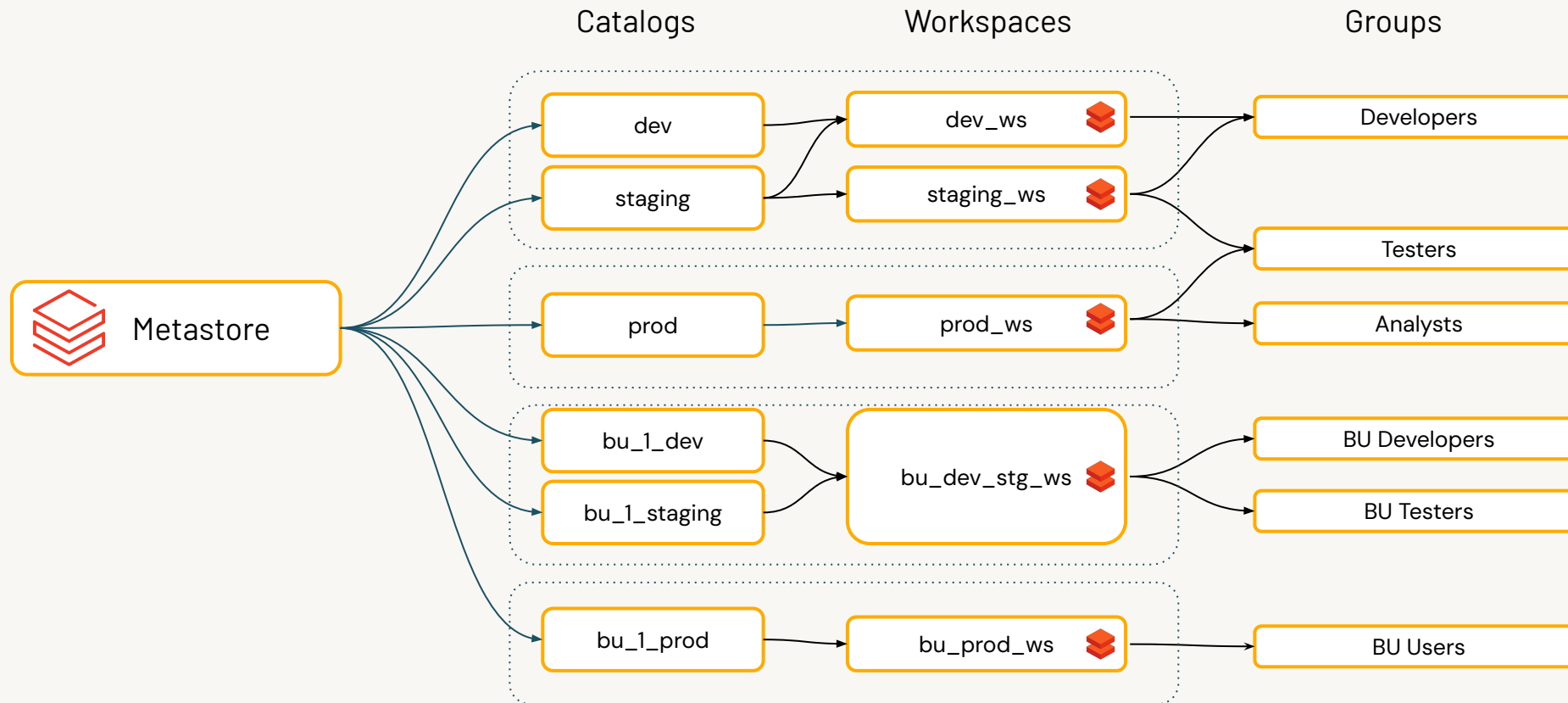
Specify mask or
function to mask

Scalable governance with ABAC



Catalog binding

Full logical and physical catalog isolation



Access to data and availability of data can be isolated across workspaces and groups

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Upgrading to Unity Catalog

Demo!

Query Federation

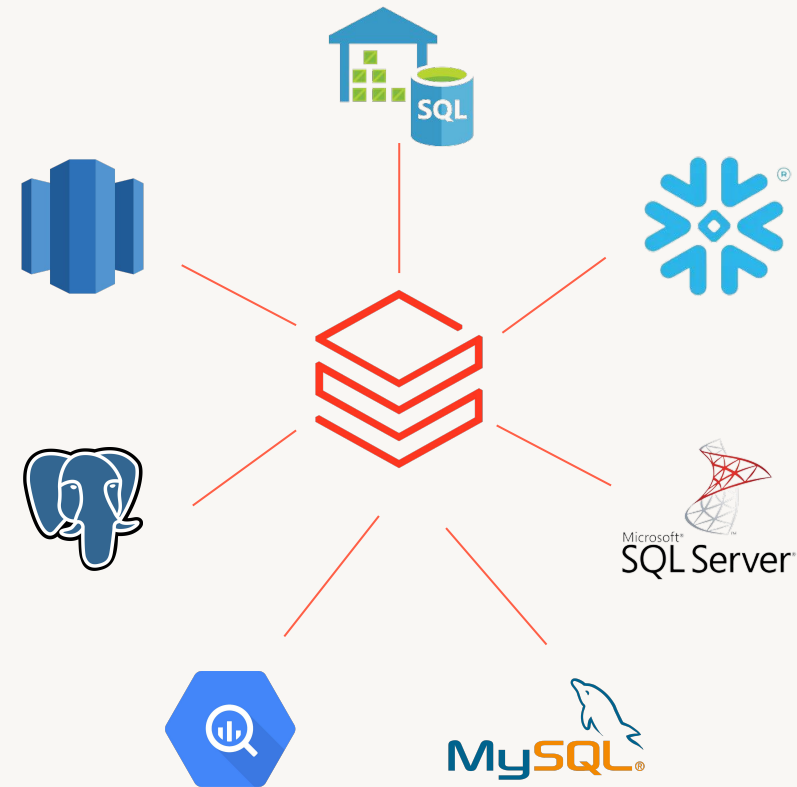
Unify your entire data estate with lakehouse

Query Federation provides **one single point of secure access to all your data** – no matter where it lives – and one way to access, catalog, govern, and query all your data – **no ingestion required**.

- **Unified permission controls**
- **Centralised credentials management**
- Intelligent pushdown optimizations
- Accelerated query performance with **Materialized Views**

```
CREATE FOREIGN CATALOG <catalog_name>  
USING CONNECTION <connection_name>  
OPTIONS (database '<remote_database>')
```

```
SELECT * FROM <catalog_name>.<schema_name>.<table_name>
```



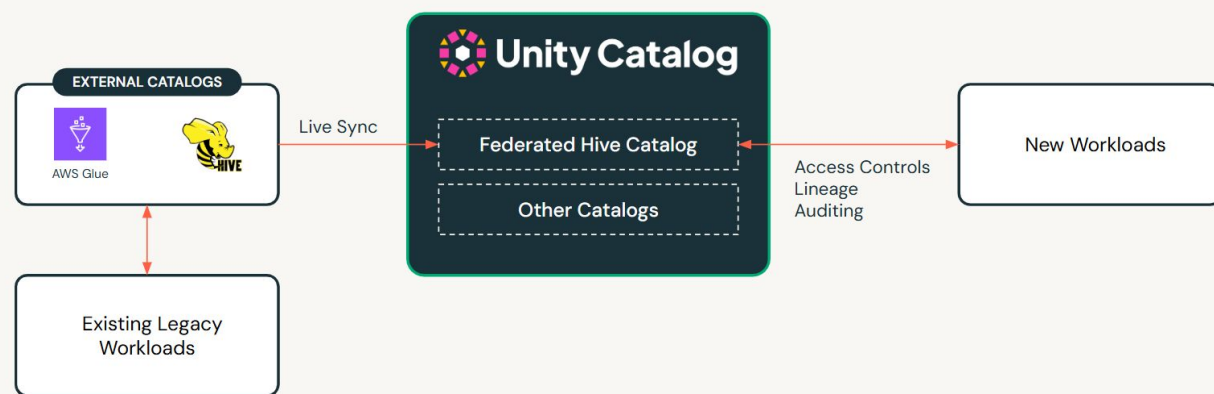
Hive Metastore and Glue Federation

Discover, govern and access data from internal/external HMS, AWS Glue

Mount internal or external Hive Metastore, AWS Glue

Access your existing assets, and apply UC governance on top for new workloads

Supports ACLs, cross workspace access, lineage, search and more



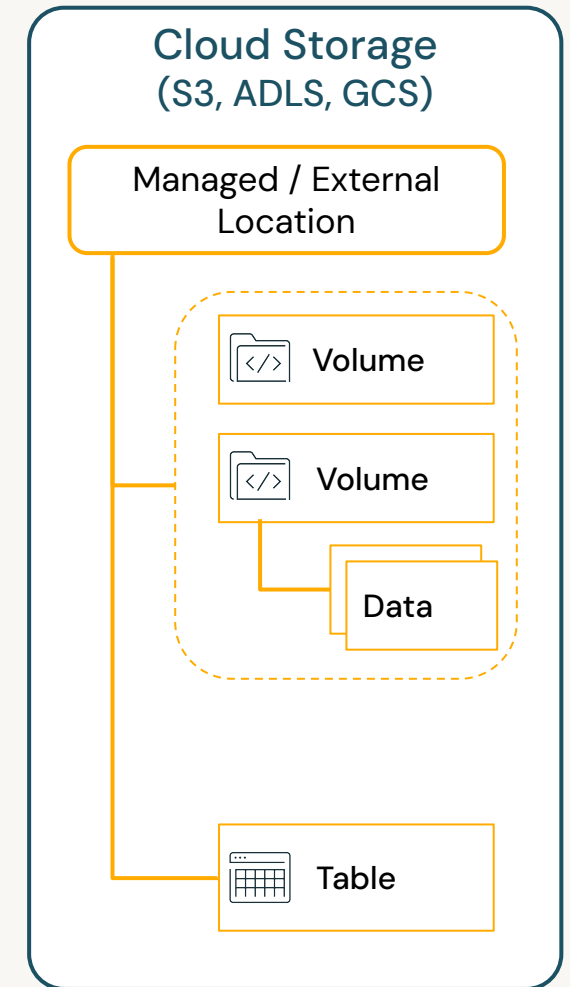
Volumes in Unity Catalog

Access, store, organize and process files with Unity Catalog governance

- Volumes can be accessed by some **POSIX commands**

```
ls /Volumes/catalog/schema/volume123
```

- Volumes are created under **Managed or External Locations** and show up in **UC Lineage**
- Volumes add governance over **non-tabular data sets**
 - Unstructured or semi-structured data, e.g., image, audio, video, or PDF files, used for ML
 - Raw data files used for ad-hoc or early stage data exploration, or saved outputs
 - Library or config files, logging or checkpointing
 -
- Tip: Tables are registered in Managed / External Locations, **not in Volumes**



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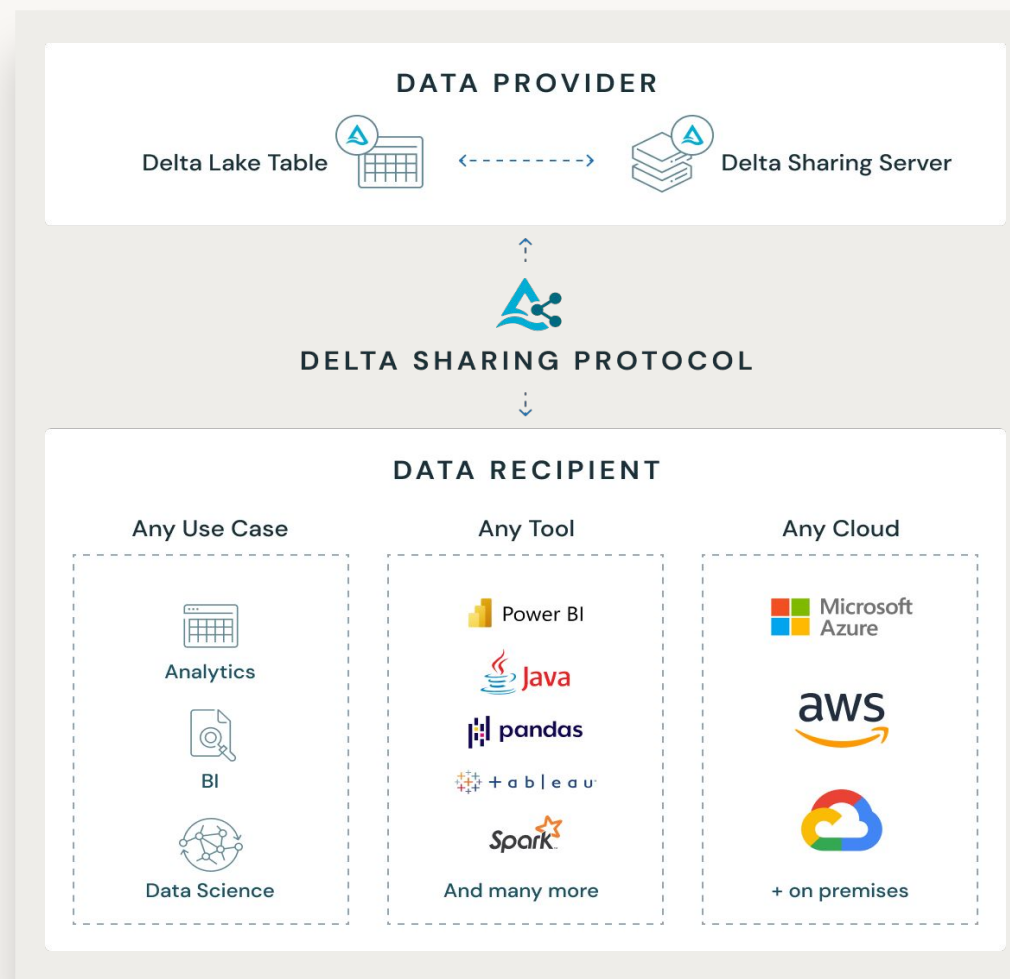
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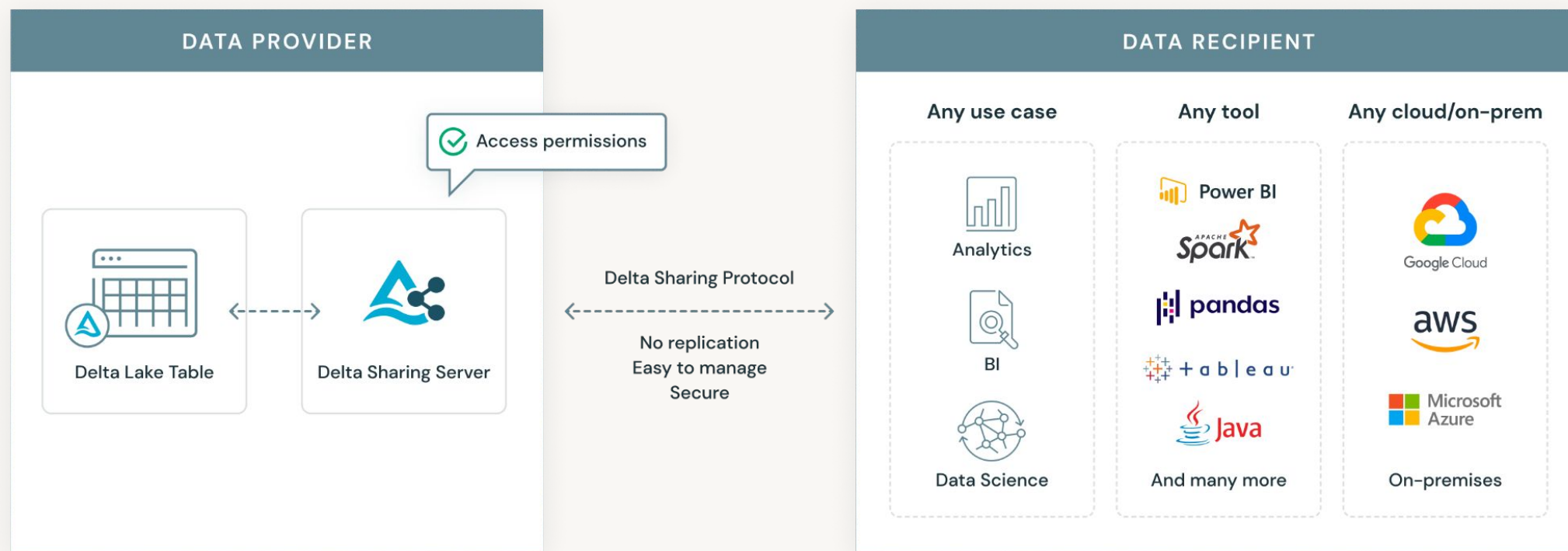
Open data sharing

- **Avoid vendor lock-in** with open source Delta Sharing for seamless data sharing across clouds, regions, and platforms, without replication
- Share **more than just data** – Notebooks, ML models, dashboards, applications
- Explore and monetize data products through an **open marketplace**
- Collaborate securely on sensitive data with **scalable data clean rooms**



A simple, open and secure approach to data sharing

Share from **Delta Sharing** to anywhere; accelerate and streamline access to data.



Databricks Marketplace

An open marketplace for data, analytics, and AI

Data sets, Notebooks, ML models and applications from top data & solution providers

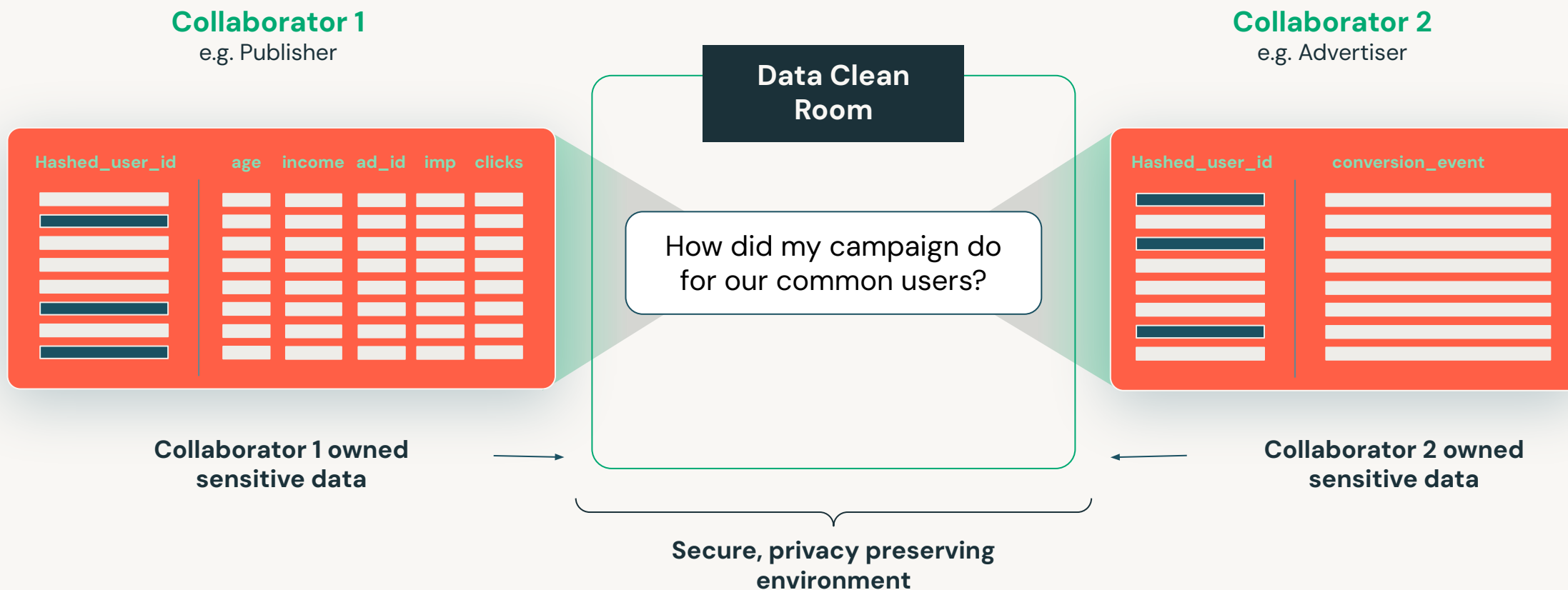
Public marketplace

Open for Databricks & non-Databricks users



Clean rooms

Secure environments to run computations on joint data



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Demo!

Why is data lineage important?

Compliance

- **Regulatory** requirements to verify data lineage
- Track the **spread of sensitive data** across datasets

Discovery

- Understand **context** and **trustworthiness** of data before using it in analytics
- **Prevent duplicative** work and data

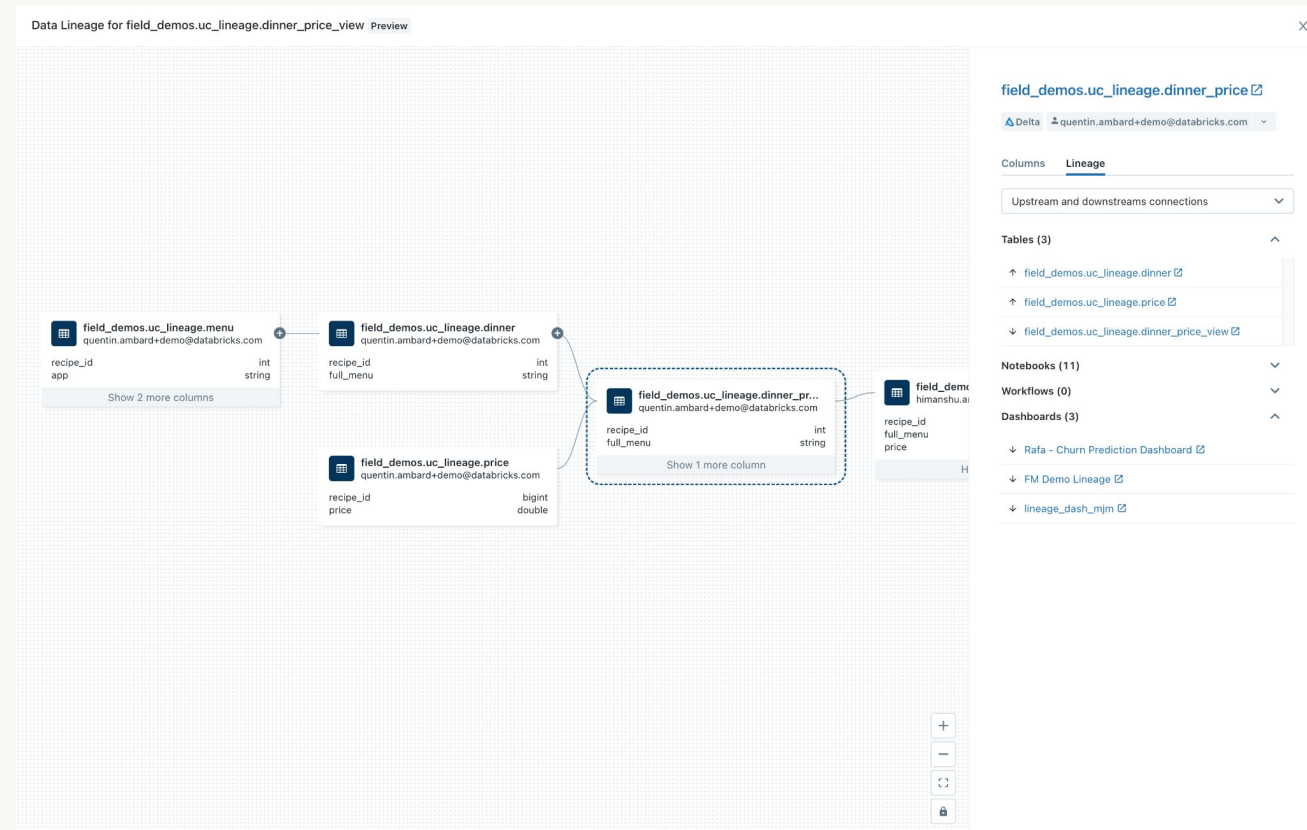
Observability

- Track down **issues / discrepancies** in reports by tracing back the data
- Analyze **impact of proposed changes** to downstream reports e.g. column deprecation

Automated lineage for all workloads

End-to-end visibility into how data flows and consumed in your organization

- **Auto-capture runtime data lineage** on a Databricks cluster or SQL warehouse
- Leverage common permission model from Unity Catalog
- Lineage across tables, columns, dashboards, workflows, notebooks, files, external sources, and models



Discovery Tags

Semantic layer for your lakehouse

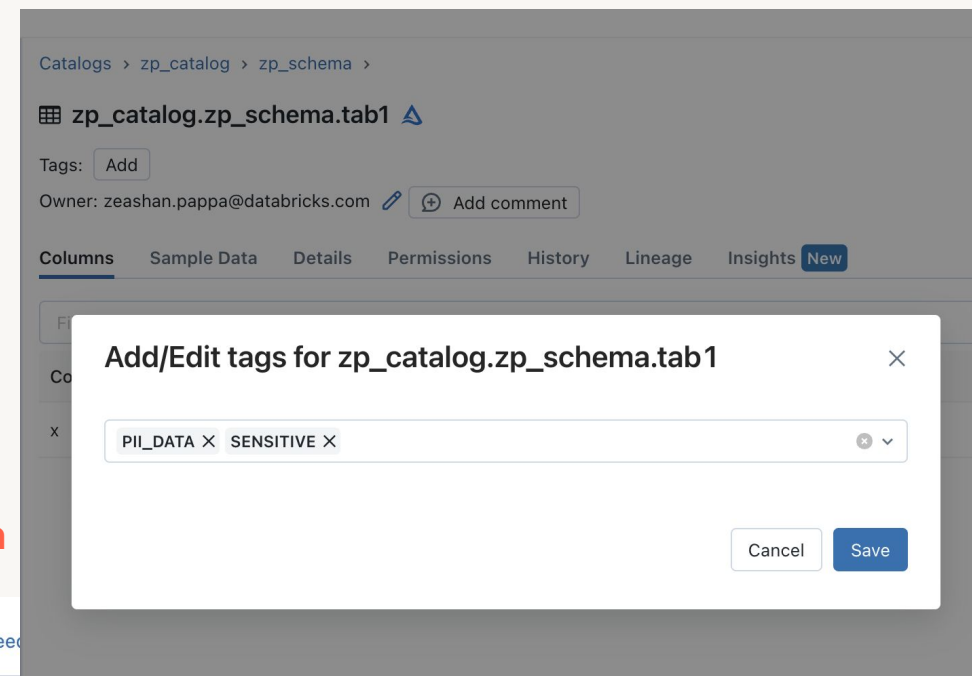
Problem

Searching for data assets in business terms or generally agreed upon taxonomies usually requires additional catalog tools.

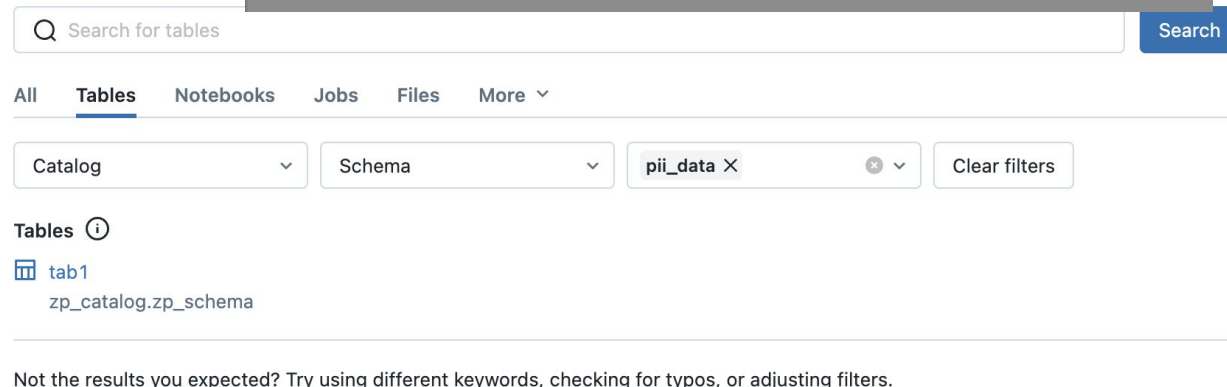
Solution

- Discovery Tags allow you to tag Column, Table, Schema, Catalog objects in UC
- **Integrated search mechanism in UC allows you to search for objects by tag.**
- Private Preview: Tag Policies (Governed Tags)

Tag your data



Search



Search based on tags

PRIVATE PREVIEW

Unity Catalog Metrics

- Enable your teams to discover, govern and reuse **reliable and consistent business metrics**
- Ensure accuracy of insights across across any databricks tool, any workload

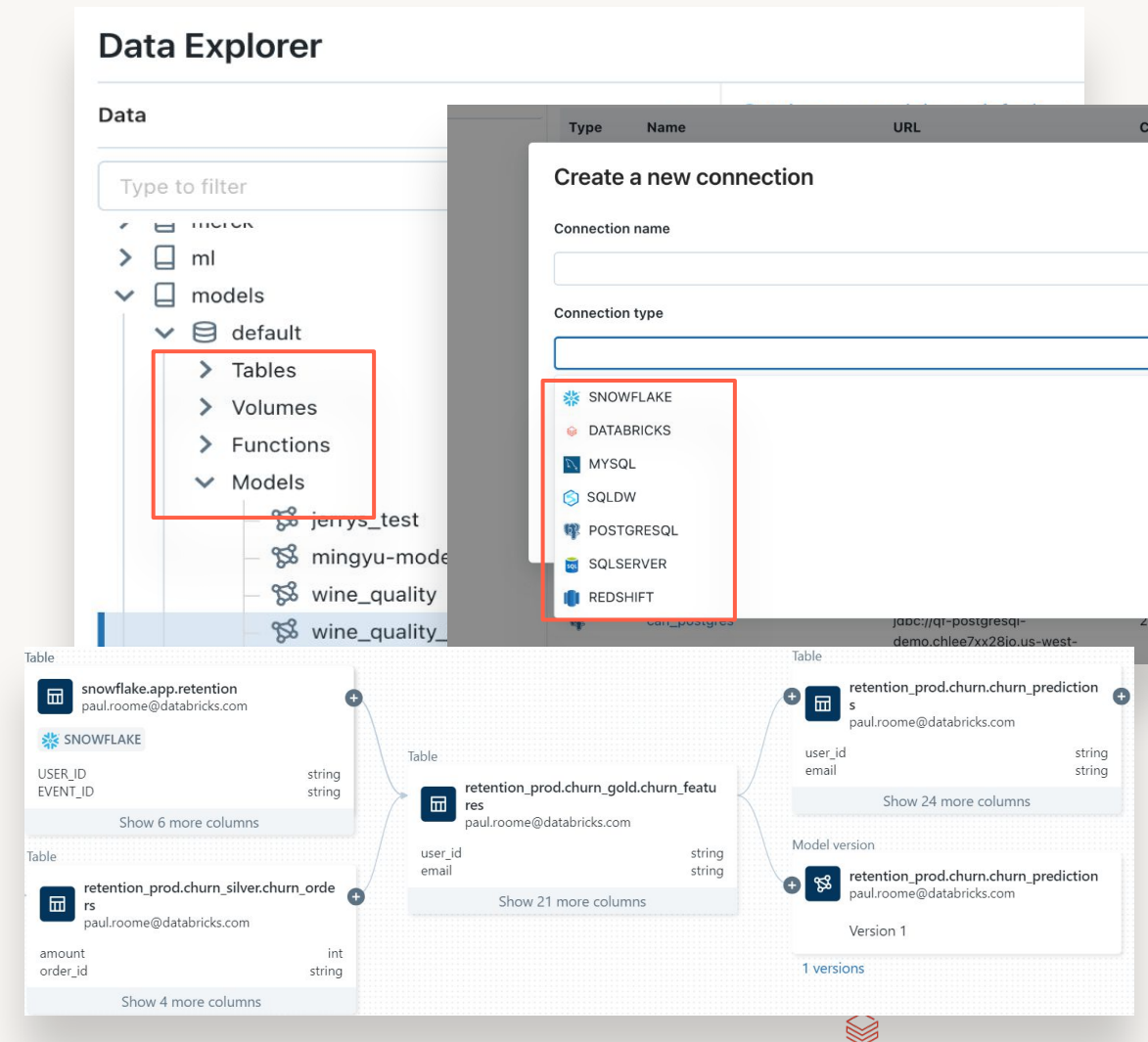


The screenshot displays the Unity Catalog interface. On the left, a sidebar shows a hierarchical view of the catalog structure, with 'semantic_sales_metrics' selected. The main panel shows the 'Overview' tab for 'semantic_sales_metrics', displaying a table of metrics with their values, trends, and types.

Measure	Value	Type
Avg Order Value basket_size Average basket size dividing total revenue by the number of net orders completed.	\$842.09 +3.1% WoW	AVG
Canceled Sales total_cancellations_revenue The number of orders or bookings that were revoked or terminated before fulfillment.	\$1.89M -2.2% WoW	SUM
COGS total_sales_cost Direct costs including logistics and marketing spend, but excludes capex and opex.	\$51.37M +0.1% WoW	SUM
Items sold items_sold Count of individual SKUs sold, includes returned SKUs as well as SKUs discarded due to quality defects.	20,345 +3.1% WoW	COUNT
Net Sales net_sales Total revenue generated, excluding sales returns, allowances, and discounts.	\$91.89M -0.7% WoW	SUM
Profit Margin profit_margin Percent of revenue that remains as profit after all cogs and sts and expenses are deducted.	+8.32% +1.4% WoW	CALC
Total Revenue total_revenue Gross income generated before any deductions or expenses are subtracted.	\$142.34M -1.8% WoW	ROLLING SUM

Unified visibility into data and AI

- **Discover and classify** structured and unstructured data, files, notebooks, ML models, and dashboards at one place
- Consolidate and query data from **other databases and data warehouses** using a **single point of access**, without moving or copying the data
- Build better **understanding of your data estate** with automated lineage, tags and auto-generated data insights
- Boost productivity by searching, understanding and gaining insights from your data and AI assets, using **natural language**



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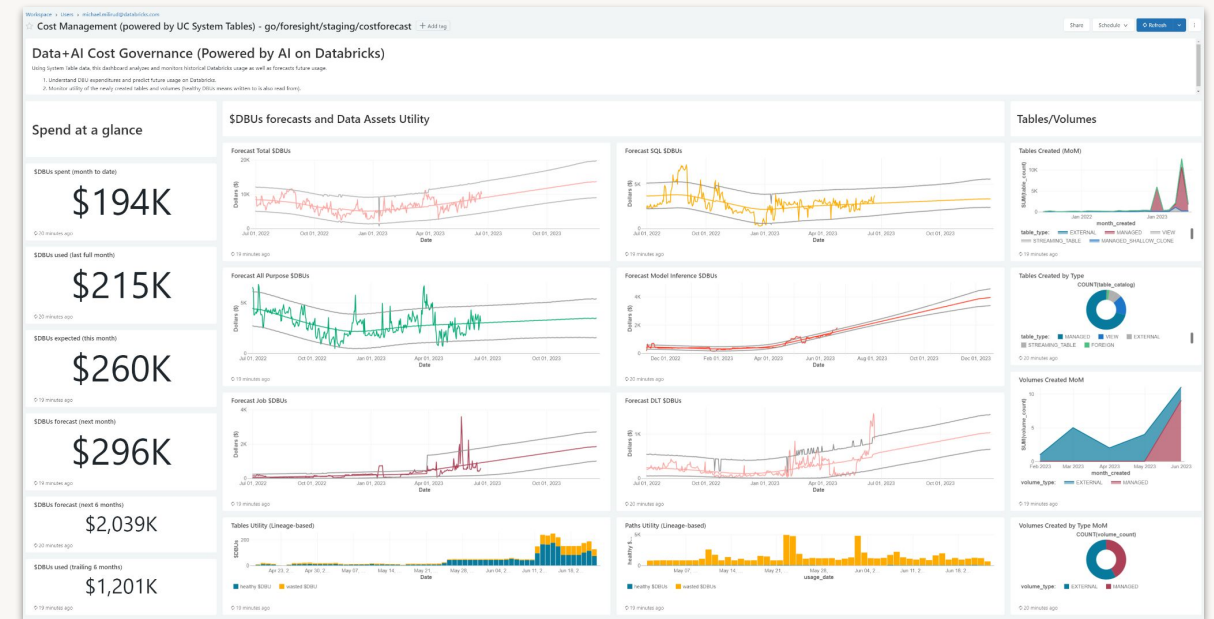
Upgrading to Unity Catalog

Demo!

System Tables

System Tables are a Databricks–hosted analytical store for **all** of Databricks operational data’s warm path used for historical customer observability, including:

- Audit/lineage analytics
- Cost/usage analytics
- Efficiency analytics
- SLO analytics
- Data Quality analytics



for jobs, notebooks, clusters, SQL/ML endpoints, etc.



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Upgrading to Unity Catalog

Demo!

Get started now!

3 ways to take advantage of Unity Catalog immediately

1. Deploy a workspace

- Deploy a workspace and leverage **Unity Catalog by default**
- Take advantage of features like comprehensive governance, lineage, AI/BI Genie Spaces, and more right away!

2. Upgrade existing Databricks workspaces

- Your Databricks team can help you **accelerate your migration** (Shared Technical services, Professional Services, Partners, ...)

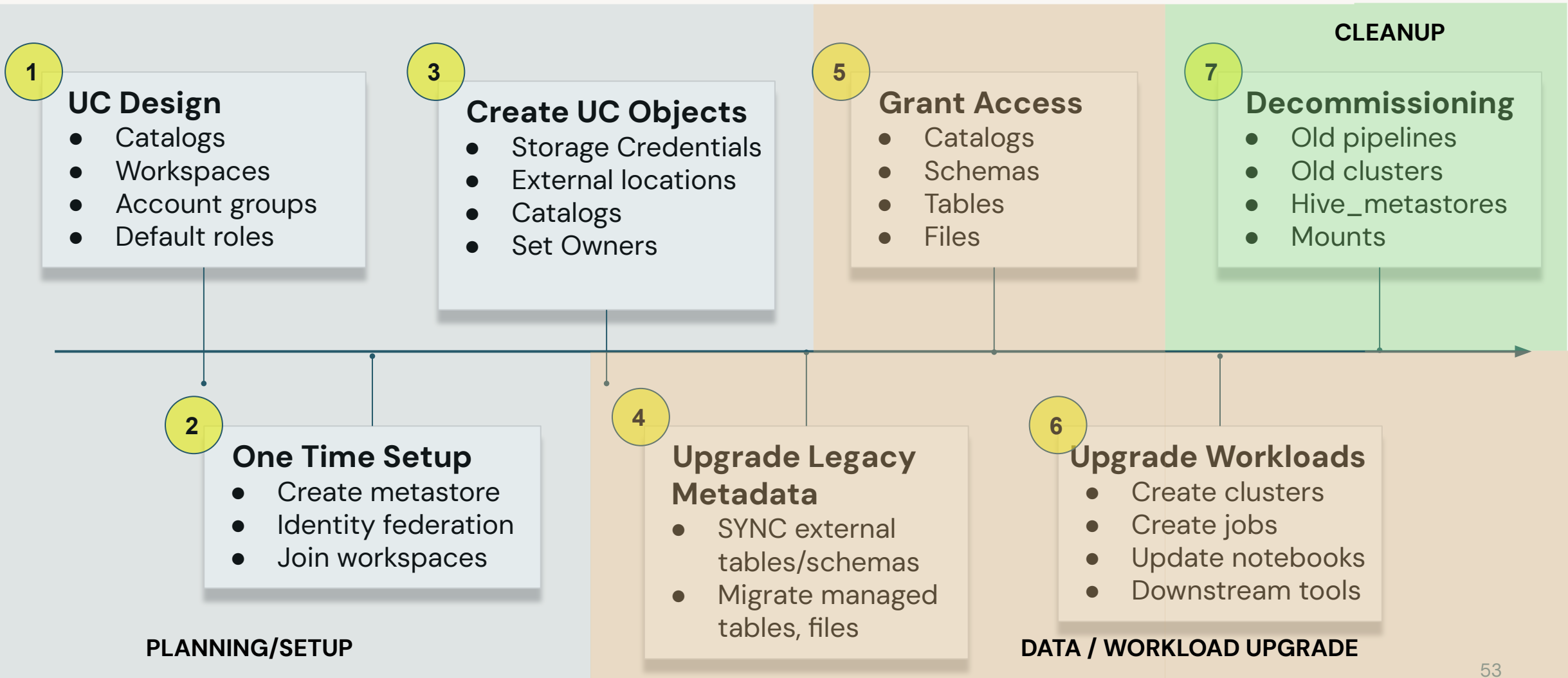
3. Use Lakehouse Federation

- **Leverage Lakehouse federation** to connect your existing data in databases, data warehouse or catalog likes HMS or Glue and get started with new workloads on Unity Catalog



How to upgrade to Unity Catalog

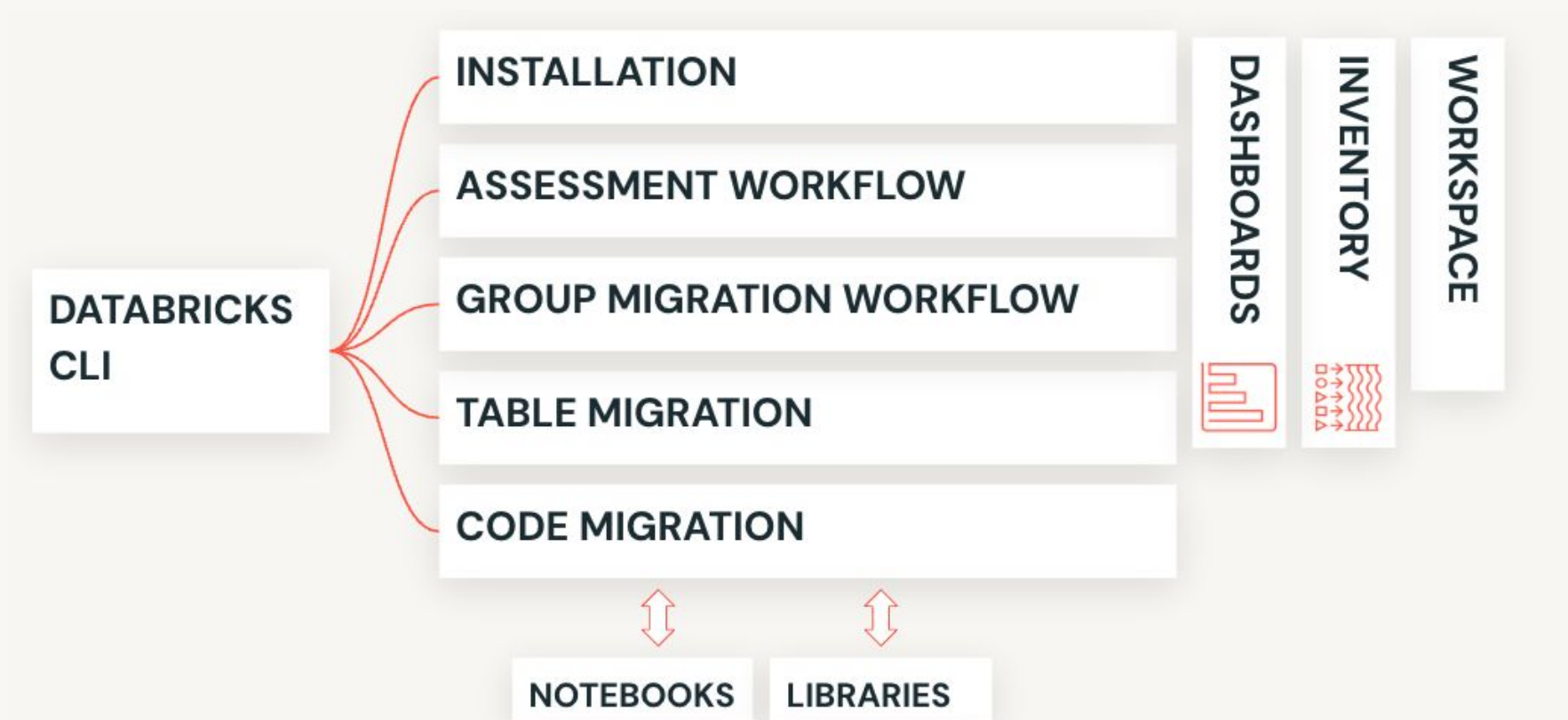
Steps to consider for a full upgrade



UCX – your best resource for UC upgrades

github.com/databrickslabs/ucx

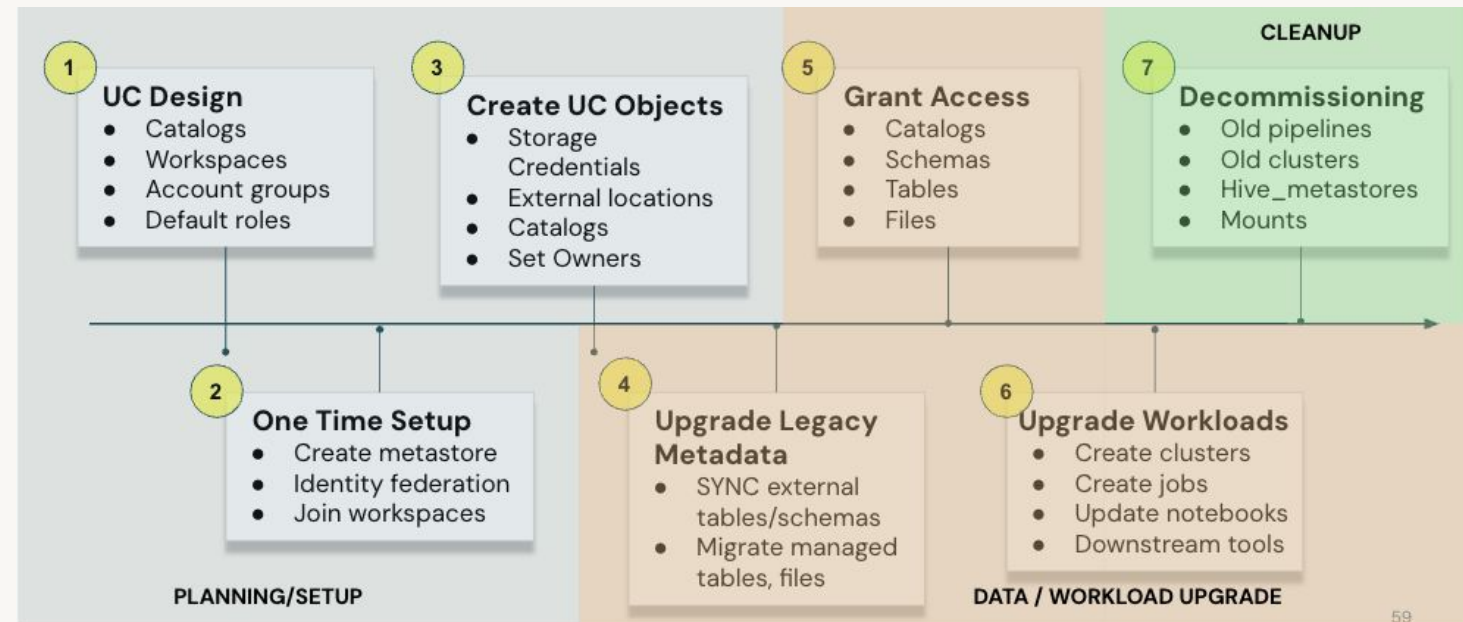
UCX is a Databricks lab project aimed at streamlining UC Adoption. It is a public source project by Databricks UC field team



How to upgrade to Unity Catalog

Steps to consider for a full upgrade

Talk to us, we are here to help!



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Demo!

Thank you!

